Rathin Gorain

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.42
Chittranjan higher secondary (EM) high school	Class XII (West Bengal Council of Higher Secondary Education)	2019	80.80%
Chittranjan X class high school	Class X (West Bengal Board of Secondary Education)	2017	78.71%

PROJECTS

Onboard Integrated Battery Charger for Multi Level Open End Winding fed EV Drives :

- **Parallel active front end converter**-based dq0 synchronous reference frame controller enabling **bidirectional (G2V/V2G)** power flow .
- Power flow near-**unity power factor** achieved by using **SRF PLL** based control technique.
- Specially designed **36-slot, 4-pole Open-End Winding** induction machine that cancels air-gap flux during charging, keeping developed torque in the **millinewton-meter range** (effectively zero)

Closed-Loop DC-DC Boost Converter -- Hardware Development & PCB Design :

- 200 W boost-converter inductor for 48-62 V in / 64-85 V out at 80 kHz, 95% efficiency target
- Designed and routed the fabrication-ready PCB using KI Cad software the converter hardware was built on,
- Implemented digital **closed-loop control** on TMS320F28049C (C2000) interfaced to the PCB

Simulink model design and analysis of Multiple output DC-DC Flyback Converter :

- Designed an isolated flyback converter with **dual regulated outputs** (5V/10W, 12V/5W) from a **200-350V DC input**, calculating duty cycle, magnetizing inductance, transformer turns ratio, and component ratings for switches and diodes.
- Simulated the converter using Simulink, achieving output voltages within 1% of design targets and verifying primary/secondary current waveforms against theoretical operation.

LED Intensity Control via ADC-PWM (TI C2000 Microcontroller) :

- Read potentiometer voltage using the on-chip 12-bit ADC and mapped it in real time to the duty cycle of a 50kHz ePWM signal,
- Achieving smooth, continuously variable LED brightness. Implemented via register-level Driver Lib programming in Code Composer Studio and verified on a DSO.

BLDC Motor Drive Simulink model design :

- Modelled 2 kW, 48 V, 3000 rpm, 8-pole BLDC motor from commercial datasheet parameters
- Designed cascaded PI speed and current control loop for closed-loop operation Implemented

SKILLS

Software : MATLAB, PLECS, Ki Cad, Altium, Ni Multisim

Programming language : C, MATLAB, Python

Hardware programming language : Verilog

RELEVANT COURSES

- Power Converter Design
- Electrical Machine Analysis & Control
- Power System Dynamics
- Steady-State Power System Analysis
- Multilevel Power Converters
- Embedded Systems for Power Electronics
- Power Electronics Laboratory
- Power Systems Laboratory
- Smart Grids
- Modeling and Advanced Control of Permanent Magnet Machines

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant (TA), Machine Laboratory** -- Assisted in conducting laboratory sessions and guided students during electrical machine experiments.
- **Teaching Assistant (TA), Power System** -- Assisted students with power-system concepts, problem-solving, and coursework.

EXTRACURRICULAR

- **Sports:** Badminton, Squash, and Cricket
- **Gaming:** Multiplayer competitive games -- Counter-Strike, BGMI, and Free Fire